

RRUHFR03

Communication Protocol v1.0



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Technical Parameters

Power

Input Voltage : 9.0 V to 30.0 V DC

Current Consumption : UPTO 3.0 A

RS232 Parameters

Baud Rate : 115200(Default)

Data Bits : 8

Parity : No Parity

Stop Bits : 1

Flow Control : No



Basic Communication Frame Format

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	XXHex
4.	Custom Request Data	X	XX...XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	XXHex
3.	Command Code	1	XXHex
4.	Error Code	1	XXHex
5.	Custom Response Data	X	XX...XXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

SOF : SOF is known as Start of frame. The byte BBHex should be transmitted in the starting of the frame.

Request / Response Frame Length: This field helps identify total bytes being transferred between the module and the host. For a request, all the bytes starting from "Command Code" to "Custom Request Data" field are included in calculation of Request Frame Length. Similarly, Response Frame Length has the count of all bytes between "Command Code" and "Custom Response Data", including both these fields.

Command Code: One byte command code is used to identify the transaction that is required to be performed by the module.

Error Code: Error code is included in all response frames and are used to identify the exact cause of a failed transaction. Two main values of Error Code are 00Hex = Success (No Error) and F0Hex = Generic Error.



Custom Request / Response Data: This field contains data that is related to the specific request or response being transferred between the module and the host. In case any specific field spans over multiple bytes, the same shall be always transmitted in MSByte first form.

Cyclic Redundancy Check (CRC): CRC field should be used to validate the data frame, either a request or a response. It ensures that the data being transferred between the module and the host is intact and valid. The value of CRC is calculated over the data starting from the "Command Code" field up to "Custom Request/Response Data" field. Being a 2 byte field, it will always be transmitted in MSByte first form. The logic to calculate the CRC value is given below.

```
unsigned int CalculateCRC()
{
    unsigned int crc = 0x0000FFFF; //32 bit unsigned int
    unsigned char bytes, bits; // 8 bit unsigned int

    for (bytes = 2; bytes < txBuffer[1] + 2; bytes++)
    {
        crc = crc ^ txBuffer[bytes];
        for (bits = 0; bits < 8; bits++)
        {
            if ((crc & 0x8000) == 0x8000)
                crc = (crc << 1) ^ 0x1021;
            else
                crc = (crc << 1);
        }
    }
    crc = (~crc);
}
```

Following is a usage example of the CalculateCRC method.

```
void main()
{
    unsigned char txBuf[256]; // 8 bit unsigned int
    unsigned int crc;

    txBuf[0] = 0xBB; // SOF
    txBuf[1] = 0x01; // Request Length
    txBuf[2] = 0x01; // Command Code
    crc = CalculateCRC();
    txBuf[3] = (unsigned char)((crc >> 8) & 0xFF);
    txBuf[4] = (unsigned char)(crc & 0xFF);

    // Request frame is ready
    // Same CRC calculation logic applies to response frame also
}
```



Command Frames

Get Device Information

This packet is used to get the information about the device.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	01Hex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	19Hex
3.	Command Code	1	01Hex
4.	Error Code	1	XXHex
5.	Firmware Version	2	XXXXHex
6.	Hardware Version	2	XXXXHex
7.	Info Length	1	XXHex
8.	Serial No	4	XX..XXHex
9.	Model Name	10	XX..XXHex
10.	Magic Number	4	XX..XXHex
11.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Hardware Version: PCB Revision - (e.g.=The value 14Hex = 20 denotes the hardware version is 2.0 (20 / 10 = 2.0))

Firmware Version: Firmware Revision - (e.g.= The value 14Hex = 20 denotes the firmware version is 2.0 (20 / 10 = 2.0))

Serial Number: It is the 4 byte Serial number of the device.

Model Name: Possible values: 52525548465230337633Hex = RRUHFR03v3



Set Frequency

This packet is used to change the frequency transmitted by the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	04Hex
3.	Command Code	1	15Hex
4.	Region	1	XXHex
5.	Max Frequency	1	XXHex
6.	Min Frequency	1	XXHex
7.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	15Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Max / Min Frequency: Possible value ranges between 0 and 9, including both these numbers. The following table explains this parameter in detail.

Min / Max Frequency Value	0	1	2	3
Actual Frequency (in KHz)	865100	865700	866300	866900



Region: Possible value ranges between 00Hex and 16Hex. The following table explains this parameter in detail.

Region Data In Hex	0	1	2	3	4	5	6	7	8
Actual Region Name	FCC	HK	TAIWAN	ETSI LOWER	ETSI UPPER	KOREA	MALAYSIA	CHINA	SOUTH AFRICA

Region Data In Hex	9	A	B	C	D	E	F
Actual Region Name	BRAZIL	THAILAND	SINGAPORE	AUSTRALIA	INDIA	URUGUAY	VIETNAM

Region Data In Hex	10	11	12	13	14	15	16
Actual Region Name	ISRAEL	PHILLIPIANS	INDONASIA	NEW ZEALAND	JAPAN 2	PERU	RUSSIA



Get Frequency

This packet is used to get the current frequency of the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	14Hex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	05Hex
3.	Command Code	1	14Hex
4.	Error Code	1	XXHex
5.	Region	1	XXHex
6.	Max Frequency	1	XXHex
7.	Min Frequency	1	XXHex
8.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Max / Min Frequency: Possible value ranges between 0 and 9, including both these numbers. The following table explains this parameter in detail.

Min / Max Frequency Value	0	1	2	3
Actual Frequency (in KHz)	865100	865700	866300	866900



Region: Possible value ranges between 00Hex and 16Hex. The following table explains this parameter in detail.

Region Data In Hex	0	1	2	3	4	5	6	7	8
Actual Region Name	FCC	HK	TAIWAN	ETSI LOWER	ETSI UPPER	KOREA	MALAYSIA	CHINA	SOUTH AFRICA

Region Data In Hex	9	A	B	C	D	E	F
Actual Region Name	BRAZIL	THAILAND	SINGAPORE	AUSTRALIA	INDIA	URUGUAY	VIETNAM

Region Data In Hex	10	11	12	13	14	15	16
Actual Region Name	ISRAEL	PHILLIPIANS	INDONASIA	NEW ZEALAND	JAPAN 2	PERU	RUSSIA



Set Link Mode

This packet is used to set the link mode in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	03Hex
3.	Command Code	1	13Hex
4.	Link Mode	2	XXXXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	13Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex



Get Link Mode

This packet is used to get the link mode in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	12Hex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	04Hex
3.	Command Code	1	12Hex
4.	Error Code	1	XXHex
5.	Link Mode	2	XXXXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex



Set Power

This packet is used to change the frequency transmitted by the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	03Hex
3.	Command Code	1	11Hex
4.	RF Output Power	2	XXXXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	11Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

RF Output Power: It denotes the output RF power value.



Get Power

This packet is used to get the frequency transmitted by the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	11Hex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	11Hex
4.	Error Code	1	XXHex
5.	RF Output Power	2	XXXXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

RF Output Power: It denotes the output RF power value.



Set Baud Rate

This packet is used to set the baud rate for RS232 in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	03Hex
3.	Command Code	1	05Hex
4.	Sub Command	1	01Hex
5.	Baud Rate Index	1	XXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	05Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Baud Rate Index: It denotes the Baud rate as below:

00Hex = 1200
 01Hex = 2400
 02Hex = 4800
 03Hex = 9600
 04Hex = 19200
 05Hex = 38400
 06Hex = 57600
 07Hex = 115200
 08Hex = 128000
 09Hex = 256000
 0AHex = 921600



Get Device Baud Rate

This packet is used to get the frequency transmitted by the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	02Hex
3.	Command Code	1	04Hex
4.	Sub Command	1	01Hex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	04Hex
3.	Command Code	1	04Hex
4.	Error Code	1	XXHex
5.	Sub Command	1	01Hex
6.	Baud Rate Index	1	XXHex
7.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Baud Rate Index: It denotes the Baud rate as below:

00Hex = 1200
 01Hex = 2400
 02Hex = 4800
 03Hex = 9600
 04Hex = 19200
 05Hex = 38400
 06Hex = 57600
 07Hex = 115200
 08Hex = 128000
 09Hex = 256000
 0AHex = 921600



Set TCP/IP Configuration

This packet is used to change the TCP/IP configurations of the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	17Hex
3.	Command Code	1	05Hex
4.	Sub Command	1	02Hex
5.	Configuration Flag	1	XXHex
5.	Device IP	4	XX..XXHex
6.	Gateway IP	4	XX..XXHex
7.	Net Mask	4	XX..XXHex
8.	Server Port	2	XXXXHex
9.	Client IP	4	XX..XXHex
10.	Client Port	2	XXHex
11.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	05Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Configuration Flag :

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
RFU	MAC Address	Client Port	Client IP	Server Port	Net Mask	Gateway	Device IP



Device IP : Contains Device IP address to set.
Gateway IP: Contains Gateway IP address to set.
Net Mask: Contains Net mask address to set.
Server Port: Contains Server port to set.
Client IP : Contains Client IP address to set.
Client Port: Contains Client port to set.
MAC ID: Contains device MAC address to set.



Get TCP/IP Configuration

This packet is used to get the current frequency of the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	02Hex
3.	Command Code	1	04Hex
4.	Sub Command	1	02Hex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	1EHex
3.	Command Code	1	04Hex
4.	Error Code	1	XXHex
5.	Sub Command	1	02Hex
6.	Configuration Flag	1	XXHex
7.	Device IP	4	XX..XXHex
8.	Gateway IP	4	XX..XXHex
9.	Net Mask	4	XX..XXHex
10.	Server Port	2	XXXXHex
11.	Client IP	4	XX..XXHex
12.	Client Port	2	XXHex
13.	MAC ID	6	XX..XXHex
14.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Configuration Flag :

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
RFU	MAC Address	Client Port	Client IP	Server Port	Net Mask	Gateway	Device IP



Device IP : Contains previously set IP address.
Gateway IP: Contains previously set gateway address.
Net Mask: Contains previously set net mask address.
Server Port: Contains previously set server port
Client IP : Contains previously set IP address.
Client Port: Contains previously set client port.
MAC ID: Contains previously set device MAC address.



Buzzer Beep

This packet is used to beep the buzzer for a specific duration in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	03Hex
3.	Command Code	1	07Hex
4.	Duration for beep buzzer	2	XXXXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	07Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Duration for beep buzzer: The duration of the buzzer is how long the buzzer should beep. The duration of the buzzer will be in multiples of 100 milliseconds.



Enable/Disable Buzzer

This packet is used to enable/disable the buzzer in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	02Hex
3.	Command Code	1	06Hex
4.	Buzzer Status	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	06Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Buzzer Status: It denotes the status of a buzzer. Possible value ranges between 00Hex = for disable buzzer and 01Hex for enable buzzer.



Device Restart

This packet is used to restart the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	0cHex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	0CHex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex



Force Boot Mode

This packet is used for firmware updates in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	0DHex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	0DHex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex



Heartbeat Setup

This packet is used to set the heartbeat time duration in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	04Hex
3.	Command Code	1	0AHex
4.	Sub Command	1	01Hex
5.	Duration of Heartbeat	2	XXXXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	0AHex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Duration for Heartbeat: The duration of the heartbeat is how long the heartbeat frame should be transmitted. The duration of the heartbeat will be in multiples of 100 milliseconds.



Set Mode of Operation

Based on the parameter provided, this command can either put the module in auto inventory mode or take out from the auto inventory mode.

While in auto inventory mode, the module will continuously check for tags and will keep sending the EPC of all the tags that are detected, without the need to send any additional command from the host. Whereas in response mode, the module will be idle until a command is received from the host. In this mode, the module will execute the operation after receiving a command from the host and respond back with the requested details.

Once a specific mode is configured into the module using this command, it will be stored inside the module. So in the case of a power cycle, the module will automatically switch to the previously configured mode.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	02Hex
3.	Command Code	1	03Hex
4.	Mode of Operation	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	03Hex
4.	Error Code	1	XXHex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Mode of Operation: Possible values: 00Hex = Response mode
01Hex = Auto inventory mode



Get Mode of Operation

This command can be used to get the current mode of operation.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	02Hex
3.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	03Hex
3.	Command Code	1	02Hex
4.	Error Code	1	XXHex
5.	Mode of Operation	1	XXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Mode of Operation: Possible values: 00Hex = Response mode
01Hex = Auto inventory mode

NOTE: If any Error occurs, then the mode of operation field will be absent.



Set Q value for Inventory

This packet is used to set the Q value for Inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	02Hex
3.	Command Code	1	17Hex
4.	QValue	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	17Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Q value: possible values: 1 to 16



Get Q value for Inventory

This packet is used to Get the Q value for Inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	16Hex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	16Hex
4.	Error Code	1	XXHex
5.	QValue	1	XXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Q value: possible values: 1 to 16

NOTE: If any Error occurs, then the Q value field will be absent.



Set Session value for Inventory

This packet is used to set the Session value for Inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	02Hex
3.	Command Code	1	19Hex
4.	Session Value	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	19Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Session value: possible values: 0 to 3



Get Session value for Inventory

This packet is used to get the Session value for Inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	18Hex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	18Hex
4.	Error Code	1	XXHex
5.	Session Value	1	XXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Session value: possible values: 0 to 3

NOTE: If any Error occurs, then the session value field will be absent.

This packet is used to set/Change the Target value for Inventory in the module.

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	02Hex
3.	Command Code	1	1BHex
4.	Target Value	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	1BHex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Target value:

Possible values: A = 00Hex
B = 01Hex
A<->B = 02Hex



Get Target value for Inventory

This packet is used to get the Target value for Inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	1AHex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	1AHex
4.	Error Code	1	XXHex
5.	Target Value	1	XXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Target value: Possible values: A = 00Hex
 B = 01Hex
 A<->B = 02Hex

NOTE: If any Error occurs, then the target value field will be absent.



Inventory(Query)

This operation is used to get the EPC value of all the tags that are available in the read zone of the RFID module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	20Hex
3.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

In the response of Inventory there will be two frames will be received.

One frame will be received for Inventory data and another frame will be received for summary.

If any Error occurs or tag is not present in the device range, then the Data frame will be absent.

Response : Data Frame

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	XXHex
3.	Command Code	1	E0Hex
4.	Error Code	1	XXHex
5.	Tag Count	1	XXHex
5.	EPC Length*	1	XXHex
6.	EPC Value*	X	XX...XXHex
7.	Cyclic Redundancy Check (CRC)	2	XXXXHex

**WHERE**

Tag Count: Count of tags detected and shared by the module

EPC Length: EPC length of respective tag in number of bytes.

EPC Value: EPC value of respective tag.

* EPC Length and EPC Value fields will be repeated for the total tags detected by the module

Response : Summary Frame

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	03Hex
3.	Command Code	1	20Hex
4.	Error Code	1	XXHex
5.	Tag Count	1	XXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Error Code: Additional value: B1Hex = No tag found
Success value : 00Hex = Successful operation

Tag Count: Count of tags detected and shared by the module

Note: If any Error occurs as mentioned above then field 5 will be absent.



Extended - Inventory

This operation is used to get the EPC, TID and RSSI values of all the tags that are available in the read zone of the RFID module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	02Hex
3.	Command Code	1	21Hex
4.	Inventory Flags	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Inventory Flags:

Bit 2 (see the table below for reference) can be configured to enable / disable RSSI value in the auto inventory response frame. All the other bits are RFU and shall always be 0.

Bit 3 (see the table below for reference) can be configured to enable / disable TID + EPC value in the auto inventory response frame. All the other bits are RFU and shall always be 0.

Bit 7 (see the table below for reference) can be configured to enable / disable TID value in the auto inventory response frame. All the other bits are RFU and shall always be 0.

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
TID	RFU	RFU	RFU	TID + EPC	RSSI	RFU	RFU

Response

In the response of Inventory there will be two frames will be received.

One frame will be received for Inventory data and another frame will be received for summary.

If any Error occurs or tag is not present in the device range, then the Data frame will be absent.

**Response : Data Frame**

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	XXHex
3.	Command Code	1	E0Hex
4.	Error Code	1	XXHex
5.	Tag Count	1	XXHex
5.	EPC Length*	1	XXHex
6.	EPC Value*	X	XX...XXHex
7.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Error Code: Additional value: 01FBHex = No tag found

Tag Count: Count of tags detected and shared by the module

EPC Length: EPC length of respective tag in number of bytes.

EPC Value: EPC value of respective tag.

* EPC Length and EPC Value fields will be repeated for the total tags detected by the module

Response : Summary Frame

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	03Hex
3.	Command Code	1	20Hex
4.	Error Code	1	XXHex
5.	Tag Count	1	XXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Error Code: Additional value: B1Hex = No tag found
Success value : 00Hex = Successful operation

Tag Count: Count of tags detected and shared by the module

Note: If any Error occurs as mentioned above then field 5 will be absent.



Read Tag Memory

This operation is used to read data from various sections of the tag memory. This command needs the EPC value of the tag for reading memory contents.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	22Hex
4.	EPC Length	1	XXHex
5.	EPC Value	X	XX...XXHex
6.	Access Password	4	XXXXXXXXHex
7.	Memory Bank	1	XXHex
8.	Start Block	2	XXXXHex
9.	Total Blocks	1	XXHex
10.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	XXHex
3.	Command Code	1	22Hex
4.	Error Code	1	XXHex
5.	Tag Memory Content	X	XX...XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Memory Type: Possible values: 00Hex = Password
 01Hex = EPC
 02Hex = TID
 03Hex = User

EPC Length: EPC length of the tag in number of bytes.

EPC Value: EPC value of the tag.



Access Password: If the tag memory is protected, correct access password value is required to read contents from the tag memory. The default value of the access password is 00000000Hex.

Start Block: The starting block number from where the data reading operation should be started. Each block contains two bytes of data. For reading EPC memory, the start block value should be 2 instead of 0.

Total Blocks: Total count of memory blocks that are required to be read from the tag. To read TID, one needs to provide value 6 to this parameter as each block contains 2 bytes of data.

Tag Memory Content: The data read from the tag based on provided details.

NOTE: If any Error occurs, then the tag memory content field will be absent.



Write Tag Memory

This operation is used to write data in various sections of the tag memory. This command needs the EPC value of the tag for writing memory contents.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	23Hex
4.	EPC Length	1	XXHex
5.	EPC Value	X	XX...XXHex
6.	Access Password	4	XXXXXXXXHex
7.	Memory Bank	1	XXHex
8.	Start Block	2	XXHex
8.	Total Blocks	1	XXHex
9.	Data to be Written	X	XX...XXHex
10.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	23Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Memory Type: Possible values: 00Hex = Password, 01Hex = EPC, 03Hex = User (TID is not included as it cannot be written)

EPC Length: EPC length of the tag in number of bytes.

EPC Value: EPC value of the tag.



Access Password: If the tag memory is protected, correct access password value is required to write contents in the tag memory. The default value of the access password is 00000000Hex.

Start Block: The starting block number from where the data writing operation should be started. Each block contains two bytes of data. For writing EPC memory, the start block value should be 2 instead of 0 as the initial two blocks in EPC are used for some other purpose.

Total Blocks: Total count of memory blocks that are required to be written in the tag.

Data to be Written: The data to be written in the tag based on specific details provided in the request.



Erase Tag Memory

This operation actually writes 00Hex value at the requested memory section of the tag. So one can also use the Write Tag Memory command with blank data value, instead of using this functionality. This operation needs the EPC value of the tag for erasing memory contents.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	24Hex
4.	EPC Length	1	XXHex
5.	EPC Value	X	XX...XXHex
6.	Access Password	4	XXXXXXXXXXHex
7.	Memory Bank	1	XXHex
8.	Start Block	2	XXXXHex
9.	Total Blocks	1	XXHex
10.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	24Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Memory Type: Possible values: 00Hex = Password, 01Hex = EPC, 03Hex = User (TID is not included as it cannot be erased)

EPC Length: EPC length of the tag in number of bytes.

EPC Value: EPC value of the tag.



Access Password: If the tag memory is protected, correct access password value is required to erase contents from the tag memory. The default value of the access password is 00000000Hex.

Start Block: The starting block number from where the data erase operation should be started. Each block contains two bytes of data. For erasing EPC memory, the start block value should be 2 instead of 0 as the initial two blocks in EPC are used for some other purpose.

Total Blocks: Total count of memory blocks that are required to be erased in the tag.



Write Tag EPC

This operation is used to change the EPC value of the tag. While using this command, please ensure that only one tag is available near the module. In case there are multiple tags, the EPC value of all of them will be changed.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
2.	Command Code	2	25Hex
3.	EPC Length	1	XXHex
4.	EPC Value	X	XX...XXHex
5.	Access Password	4	XXXXXXXXXXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	25Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

EPC Length: The required EPC length of the tag in number of bytes. Please note that the max possible length of EPC depends on the tag chip. Even though the length of EPC can be changed, it has to be in multiples of 2 bytes as one block contains 2 bytes of memory.

EPC Value: New EPC value of the tag.

Access Password: If the tag memory is protected, correct access password value is required to write contents in the tag memory. The default value of the access password is 00000000Hex.



Kill Tag

This operation is used to permanently kill the tag. Once killed, the tag will never respond to any command from the module. Kill tag is an irreversible operation.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	27Hex
4.	RFU	1	00Hex
5.	EPC Length	1	XXHex
6.	EPC Value	X	XX...XXHex
7.	Kill Password	4	XXXXXXXXXXHex
8.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	27Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

EPC Length: EPC length of the tag in number of bytes.

EPC Value: EPC value of the tag.

Kill Password: To kill a tag, a valid kill password value is required. The default value of the kill password is 00000000Hex.



Set Tag Protection

This operation is used to protect various memory sections of a tag. This command needs the EPC value of the tag for protecting the same.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	26Hex
4.	EPC Length	1	XXHex
5.	EPC Value	X	XX...XXHex
6.	Access Password	4	XXXXXXXXXXHex
7.	Memory Type	1	XXHex
8.	Protection Type	1	XXHex
9.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	26Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

EPC Length: EPC length of the tag in number of bytes.

EPC Value: EPC value of the tag.

Access Password: If the tag memory is protected, correct access password value is required to write contents in the tag memory. The default value of the access password is 00000000Hex.

Memory Type: Possible values: 01Hex = User Memory
 02Hex = TID Memory
 03Hex = EPC Memory
 04Hex = Access Password
 05Hex = Kill Password



Protection Type: Depending on the memory type, the protection level differs. Following are three different possibilities for protection type.

For Memory Type- Kill Password, Access Password and User Memory:

01Hex = Read / write from any state, 02Hex = Permanently read / write,
03Hex = Read / write from the secured state, 04Hex = Never read / write

For Memory Type- EPC Memory:

00Hex = Write from any state, 01Hex = Permanently write, 02Hex = Write
from the secured state, 03Hex = Never write

For Memory Type- TID Memory:

00Hex = Read from any state, 01Hex = Permanently read, 02Hex = Read
from the secured state, 03Hex = Never read



Relay Trigger for one shot

This packet is used to trigger the relay for a particular time period in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	04Hex
3.	Command Code	1	08Hex
4.	Relay Number	1	XXHex
5.	Duration for relay trigger	2	XXXXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	08Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Relay Number: Possible values: 03Hex = Relay1
02Hex = Relay2

Duration for relay trigger: It is the duration time for how long the relay will be triggered.



Relay On/Off

This packet is used to turn on or off the relay in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	03Hex
3.	Command Code	1	08Hex
4.	Sub Command	1	0CHex
5.	Relay Number	1	XXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	08Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Relay Number: Possible values: 04Hex = Relay 1
06Hex = Relay 2



Get Extended Auto-Inventory Configuration Flags

This packet is used to get the values of flags for extended Auto-Inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	02Hex
3.	Command Code	1	D5Hex
4.	Sub Command	1	01Hex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	D5Hex
4.	Error Code	1	XXHex
5.	Sub Command	1	01Hex
6.	Configuration Register 1	1	XXHex
7.	Configuration Register 2	1	XXHex
8.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Configuration Registers:

Config Reg 1:

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
Fail Result Output	Pass Result Output	Compliant Tags Only	User Memory	TID	RSSI	Access Password	EPC

**Config Reg 2:**

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
Inventory Operation	Reader ID	Buffered Read Mode	RFU	Heartbe at	Soft Trigger	Persistence	Input Trigger

NOTE: If any Error occurs, then the configuration Register 1 and configuration register 2 field will be absent.



Set Extended Auto-Inventory Configuration Flags

This packet is used to set the values of flags for extended Auto-Inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	06Hex
3.	Command Code	1	D5Hex
4.	Sub Command	1	02Hex
5.	Configuration Register 1	1	XXHex
6.	Configuration Register 2	1	XXHex
7.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	03Hex
3.	Command Code	1	D5Hex
4.	Error Code	1	XXHex
5.	Sub Command	1	01Hex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Configuration Registers:

Config Reg 1:

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
Fail Result Output	Pass Result Output	Compliant Tags Only	User Memory	TID	RSSI	Access Password	EPC

Config Reg 2:

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
Inventory Operation	Reader ID	Buffered Read Mode	RFU	Heartbeat	Soft Trigger	Persistence	Input Trigger



Set Tag Persistence value

This packet is used to get the Target value for Inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	05Hex
3.	Command Code	1	D8Hex
4.	Sub Command	1	02Hex
5.	Enable Flag	1	XXHex
6.	Persistence Duration	2	XXXXHex
7.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	03Hex
3.	Command Code	1	D8Hex
4.	Error Code	1	XXHex
5.	Sub Command	1	02Hex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Duration for persistence: The duration of the persistence will be in multiples of 100 milliseconds.



Get Tag Persistence value

This packet is used to get the Target value for Inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	02Hex
3.	Command Code	1	D8Hex
4.	Sub Command	1	01Hex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	06Hex
3.	Command Code	1	D8Hex
4.	Error Code	1	XXHex
5.	Sub Command	1	01Hex
6.	Enable Flag	1	XXHex
7.	Persistence Duration	2	XXXXHex
8.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

Duration for persistence: The duration of the persistence will be in multiples of 100 milliseconds.



Set Configurations for Extended Auto-Inventory

This packet is used to set the configurations for extended auto inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	D2Hex
4.	Configuration Register 1	1	XXHex
5.	Configuration Register 2	1	XXHex
6.	EPC Mask Length	1	XXHex
7.	EPC Mask Value	XX	XX..XXHex
8.	Access Password	4	XX..XXHex
9.	Block Address	2	XXXXHex
10.	Block Count	1	XXHex
11.	Pass enable I/O	1	XXHex
12.	Pass Enable active State	1	XXHex
13.	Pass Enable Duration	2	XXXXHex
14.	Fail enable I/O	1	XXHex
15.	Fail Enable active State	1	XXHex
16.	Fail Enable Duration	2	XXXXHex
17.	Trigger Input I/O	1	04Hex
18.	Trigger Input active state	1	XXHex
19.	Trigger input Duration	2	XXXXHex
20.	Persistence Duration	2	XXXXHex
21.	Heartbeat Duration	2	XXXXHex
22.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	D2Hex
4.	Error Code	1	XXHex
5.	Cyclic Redundancy Check (CRC)	2	XXXXHex



WHERE

Configuration Registers:

Config Reg 1:

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
Fail Result Output	Pass Result Output	Compliant Tags Only	User Memory	TID	RSSI	Access Password	EPC

Config Reg 2:

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
Inventory Operation	Reader ID	Buffered Read Mode	RFU	Heartbeat	Soft Trigger	Persistence	Input Trigger

EPC Mask Length: Length of the EPC mask bytes.

EPC Mask value: The tag will be considered a valid tag. Only if the tag's EPC value matches with the configured mask value.

Access Password: If the tag memory is protected, correct access password value is required to write contents in the tag memory. The default value of the access password is 00000000Hex.

Block Address: It is the starting block number from where the data for User memory read operation should be started. Each block contains two bytes of data. For Read User memory, the start block value should be 0 as the initial two blocks.

Block Count: Total count of memory blocks that are required to be read from the tag. To read TID, one needs to provide value 6 to this parameter as each block contains 2 bytes of data.

Pass enable i/o: Possible Values: 00Hex to 04Hex

Pass Enable active State: Possible values: Set = 01Hex
Reset = 00Hex

Pass Enable Duration: The duration of the Pass enable is how long the Pass Enable frame should be transmitted. The duration of the Pass enable will be in multiples of 100 milliseconds.

Fail enable i/o: Possible Values: 00Hex to 04Hex



Fail Enable active State: Possible values: Set = 01Hex
Reset = 00Hex

Fail Enable Duration: The duration of the Fail enable is how long the Fail Enable frame should be transmitted. The duration of the Fail enable will be in multiples of 100 milliseconds.

Trigger input i/o: Possible Values: 04Hex

Trigger Input active state: Possible values: Set = 01Hex
Reset = 00Hex

Trigger input Duration: The duration of the Trigger input is how long the Trigger input frame should be transmitted. The duration of the Trigger input will be in multiples of 100 milliseconds.

Persistence Duration: The duration of the Persistence is how long the Persistence frame should be transmitted. The duration of the Persistence will be in multiples of 100 milliseconds.

Heartbeat Duration: The duration of the heartbeat is how long the heartbeat frame should be transmitted. The duration of the heartbeat will be in multiples of 100 milliseconds.



Get Configurations for Extended Auto-Inventory

This packet is used to get the configurations for extended auto inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	01Hex
3.	Command Code	1	D1Hex
4.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Response Frame Length	1	02Hex
3.	Command Code	1	D1Hex
4.	Error Code	1	XXHex
5.	Configuration Register 1	1	XXHex
6.	Configuration Register 2	1	XXHex
7.	EPC Mask Length	1	XXHex
8.	EPC Mask Value	XX	XX..XXHex
9.	Access Password	4	XX..XXHex
10.	Block Address	2	XXXXHex
11.	Block Count	1	XXHex
12.	Pass enable I/O	1	XXHex
13.	Pass Enable active State	1	XXHex
14.	Pass Enable Duration	2	XXXXHex
15.	Fail enable I/O	1	XXHex
16.	Fail Enable active State	1	XXHex
17.	Fail Enable Duration	2	XXXXHex
18.	Trigger Input I/O	1	04Hex
19.	Trigger Input active state	1	XXHex
20.	Trigger input Duration	2	XXXXHex
21.	Persistence Duration	2	XXXXHex
22.	Heartbeat Duration	2	XXXXHex
23.	Cyclic Redundancy Check (CRC)	2	XXXXHex



WHERE

Configuration Registers:

Config Reg 1:

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
Fail Result Output	Pass Result Output	Compliant Tags Only	User Memory	TID	RSSI	Access Password	EPC

Config Reg 2:

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
Inventory Operation	Reader ID	Buffered Read Mode	RFU	Heartbeat	Soft Trigger	Persistence	Input Trigger

EPC Mask Length: Length of the EPC mask bytes.

EPC Mask value: The tag will be considered a valid tag. Only if the tag's EPC value matches with the configured mask value.

Access Password: If the tag memory is protected, correct access password value is required to write contents in the tag memory. The default value of the access password is 00000000Hex.

Block Address: It is the starting block number from where the data for User memory read operation should be started. Each block contains two bytes of data. For Read User memory, the start block value should be 0 as the initial two blocks.

Block Count: Total count of memory blocks that are required to be read from the tag. To read TID, one needs to provide value 6 to this parameter as each block contains 2 bytes of data.

Pass enable i/o: Possible Values: 00Hex to 04Hex

Pass Enable active State: Possible values: Set = 01Hex
Reset = 00Hex

Pass Enable Duration: The duration of the Pass enable is how long the Pass Enable frame should be transmitted. The duration of the Pass enable will be in multiples of 100 milliseconds.

Fail enable i/o: Possible Values: 00Hex to 04Hex

Fail Enable active State: Possible values: Set = 01Hex



Reset = 00Hex

Fail Enable Duration: The duration of the Fail enable is how long the Fail Enable frame should be transmitted. The duration of the Fail enable will be in multiples of 100 milliseconds.

Trigger input i/o: Possible Values: 04Hex

Trigger Input active state: Possible values: Set = 01Hex
Reset = 00Hex

Trigger input Duration: The duration of the Trigger input is how long the Trigger input frame should be transmitted. The duration of the Trigger input will be in multiples of 100 milliseconds.

Persistence Duration: The duration of the Persistence is how long the Persistence frame should be transmitted. The duration of the Persistence will be in multiples of 100 milliseconds.

Heartbeat Duration: The duration of the heartbeat is how long the heartbeat frame should be transmitted. The duration of the heartbeat will be in multiples of 100 milliseconds.



Set EPC Mask value

This packet is used to set the EPC Mask value in EPC Mask Bank for extended auto inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	D3Hex
4.	Sub Command	1	01Hex
5.	EPC Mask Bank Number	1	XXHex
6.	Mask Data Length	X	XXHex
7.	Mask Data	XX	XX..XXHex
8.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	D3Hex
4.	Error Code	1	XXHex
5.	Sub Command	1	01Hex
6.	EPC Mask Bank Number	1	XXHex
7.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

EPC Mask Bank Number: Possible values: 00Hex to 09Hex.

Mask Data Length: It denotes the length of EPC Mask Data.

Mask Data: It denotes the EPC Mask Data for EPC masking.



Get EPC Mask value

This packet is used to get the EPC Mask value in EPC Mask Bank for extended auto inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	D3Hex
4.	Sub Command	1	02Hex
5.	EPC Mask Bank Number	1	XXHex
8.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	D3Hex
4.	Error Code	1	XXHex
5.	Sub Command	1	01Hex
6.	EPC Mask Bank Number	1	XXHex
7.	EPC Mask Flag	1	XXHex
8.	RFU	1	00Hex
9.	Mask Data Length	X	XXHex
10.	Mask Data	XX	XX..XXHex
11.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

EPC Mask Bank Number: Possible values: 00Hex to 09Hex.

**EPC Mask Flag:**

Bit 7 (MSBit)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSBit)
Valid Data	RFU	RFU	RFU	RFU	RFU	RFU	Status Flag

Valid Data: The stored data in EPC Mask bank is valid or not

Status Flag: Possible value: 00 - Mask value flag disable
01 - Mask value flag enable

Mask Data Length: It denotes the length of EPC Mask Data.

Mask Data: EPC Mask Data for EPC masking.



Errase EPC Mask value

This packet is used to erase the EPC Mask value in EPC Mask Bank for extended auto inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	03Hex
3.	Command Code	1	D3Hex
4.	Sub Command	1	04Hex
5.	EPC Mask Bank Number	1	XXHex
6.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	XXHex
3.	Command Code	1	D3Hex
4.	Error Code	1	XXHex
5.	Sub Command	1	04Hex
6.	EPC Mask Bank Number	1	XXHex
7.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

EPC Mask Bank Number: Possible values: 00Hex to 09Hex.



Enable/Disable EPC Mask Flag value

This packet is used to Enable/Disable the EPC Mask Bank Flag value for particular mask bank number for extended auto inventory in the module.

Request

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	04Hex
3.	Command Code	1	D3Hex
4.	Sub Command	1	03Hex
5.	EPC Mask Bank Number	1	XXHex
6.	Status Flag	1	XXHex
7.	Cyclic Redundancy Check (CRC)	2	XXXXHex

Response

Sr. No.	Parameter	Length (in Bytes)	Data
1.	SOF	1	BBHex
2.	Request Frame Length	1	04Hex
3.	Command Code	1	D3Hex
4.	Error Code	1	XXHex
5.	Sub Command	1	03Hex
6.	EPC Mask Bank Number	1	XXHex
7.	Cyclic Redundancy Check (CRC)	2	XXXXHex

WHERE

EPC Mask Bank Number: Possible values: 00Hex to 09Hex.

Status Flag: Possible value : 00Hex - Disable EPC Mask value
01Hex - Enable EPC Mask value.



Revision History

Sr. No.	Revision Date	Details	Revised By	Approved By
1	25/12/2024	Initial Draft - v1.0	KG	DP